Sufficient Statistics

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Introduction

A statistic is **sufficient** for a parameter if it captures all the information in the data that is relevant to that parameter. Once a sufficient statistic is computed, the raw data have nothing left to add about the parameter. Sufficiency organises the theory of estimation: it explains why the sample mean is a complete summary for the normal mean, why the sum of successes suffices for the binomial proportion, and why the exponential family of distributions is central to modern statistical modelling.

Prerequisites

The reader should be comfortable with the likelihood function, conditional probability, and a little abstract notation.

Theory

Formally, $T(X_1, \ldots, X_n)$ is **sufficient for** θ if the conditional distribution of the data given $T = t$ does not depend on θ. Equivalently (Fisher-Neyman factorisation theorem), T is sufficient iff the joint density factorises as

$$f(x_1, \ldots, x_n \mid \theta) = g\big(T(x), \theta\big) \cdot h(x),$$

where h does not depend on θ and g depends on the data only through T.

Examples:

- **Normal** (μ, σ^2) **with known** σ^2: $T = \bar{X}$ is sufficient for μ.
- **Normal with both unknown**: $(\bar{X}, \sum(X_i - \bar{X})^2)$ is jointly sufficient.
- **Bernoulli/Binomial**: $T = \sum X_i$ is sufficient for π.
- **Poisson**: $T = \sum X_i$ is sufficient for λ.
- **Uniform**$(0, \theta)$: $T = \max X_i$ is sufficient for θ.

A statistic is **minimal sufficient** if it is a function of every other sufficient statistic – i.e., it is the most compact sufficient summary.

The **exponential family** has density $f(x \mid \theta) = h(x) \exp[\eta(\theta)^\top T(x) - A(\theta)]$ for natural statistic $T(x)$. Almost every distribution used in applied statistics (normal, binomial, Poisson, gamma, beta, multinomial) belongs to the exponential family, and $T(x)$ is minimal sufficient for $\eta(\theta)$.

Rao-Blackwell theorem: given any unbiased estimator $\hat{\theta}$ and a sufficient statistic T, the conditional expectation $E[\hat{\theta} \mid T]$ is also unbiased and has variance no larger than $\hat{\theta}$. Conditioning on sufficient statistics never hurts and may help.

Assumptions

The factorisation theorem applies to proper densities on a common support. In non-regular families (uniform with an unknown upper endpoint) sufficiency still applies, but the resulting estimators inherit the non-standard behaviour of the family.

R Implementation

```r
set.seed(2026)

n <- 50
mu <- 5
sigma <- 2
x <- rnorm(n, mean = mu, sd = sigma)

T1 <- sum(x)
T2 <- sum(x^2)

log_lik_from_sum <- function(mu, sigma, T1, T2, n) {
  -n/2 * log(2 * pi * sigma^2) -
    (T2 - 2 * mu * T1 + n * mu^2) / (2 * sigma^2)
}

mu_hat <- T1 / n
sigma2_hat <- (T2 - T1^2/n) / n

c(mu_hat = mu_hat, sigma2_hat = sigma2_hat)

c(mle_from_data_mean = mean(x),
  mle_from_data_var  = mean((x - mean(x))^2))
```

Computing the MLE from only the two sufficient statistics $(T_1, T_2) = (\sum x, \sum x^2)$ yields the same point estimates as computing it from the full data vector.

Output & Results

```
    mu_hat sigma2_hat
     5.090      3.821

mle_from_data_mean  mle_from_data_var
             5.090              3.821
```

The two are numerically identical, confirming that (T_1, T_2) contains all the information about the parameters that the data can provide.

Interpretation

In practice, sufficiency is used under the hood. When a software package reports a confidence interval for a normal mean using `t.test()`, it relies on the fact that $\bar{X}$ and s^2 are sufficient for (μ, σ^2). For a manuscript, sufficiency is rarely mentioned explicitly but justifies the use of summary statistics as complete descriptions of the data in the parametric model assumed.

Practical Tips

- Sufficient statistics let you store compressed summaries (counts, sums) without losing information for inference – useful for distributed computation.
- The exponential family is the most important family to understand, because almost every GLM assumes it.
- Completeness plus sufficiency plus unbiasedness implies the UMVUE (Lehmann-Scheffe); many textbook UMVUE proofs rely on this.

- Rao-Blackwellisation is useful in MCMC: conditioning on sufficient statistics for part of the parameter vector reduces Monte Carlo variance.
- Sufficiency depends on the model. A statistic that is sufficient for μ in a normal model may carry no information about the same μ under a different data-generating process.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Scientific process and research workflow
- R, RStudio, Quarto, and renv toolchain
- Data types, tidy data, accuracy and precision
- Import, joins, and missingness with dplyr